Hw 5

Multiple of 3, b/w 0 and 12

2a) $\{0, 3, 6, 9, 12\}$

$\{x \in \mathbb{Z} \mid 0 \leq x \leq 12, \text{ and } 3 \text{ divides } x\}$

or

$\{x \in \mathbb{Z} \mid x = 3n \text{ for some } n \in \{0, 1, 2, 3, 4\}\}$

$\{x \in \mathbb{Z} \mid x = 3n, 0 \leq n \leq 4\}$ ✓

2a) A = set of sophomores at Maryland

B = set of students in discrete mathematics at Maryland

✓ $A \cap B$ = set of students taking discrete mathematics at Maryland

40. $(A \oplus B) \oplus C = A \oplus (B \oplus C)$

$(A \oplus B) = (A - B) \cup (B - A)$ → minus specifies "not"

$(A \oplus B) = (A \cup B) - (A \cap B)$ Venn Diagram

equivalent ✓

Truth Table:

A	B	C	$A \oplus B$	$(A \oplus B) \oplus C$	$B \oplus C$	$A \oplus (B \oplus C)$
T	T	T	F	T	F	T
T	T	F	F	F	T	F
T	F	F	T	T	F	T
F	T	T	T	F	F	F
F	F	T	F	T	T	T
F	T	F	T	T	T	T
T	F	T	T	F	T	F
F	F	F	F	F	F	F

$(A \oplus B) \oplus C = A \oplus (B \oplus C)$ ✓ → The columns associated with these 2 expressions are the same, meaning the overall expression is valid by associativity.